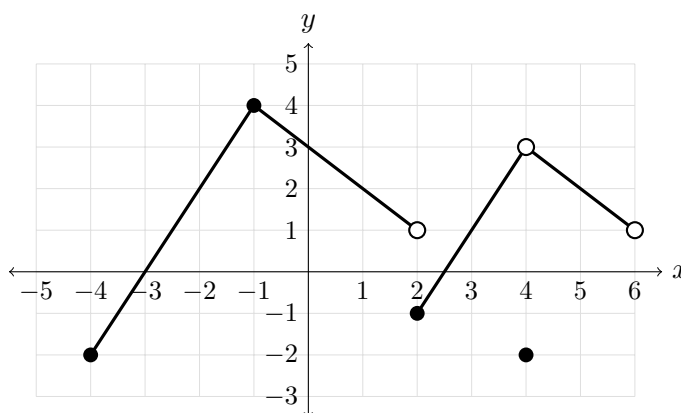


Midterm 1 Review Sheet

Math 133: Business Calculus

Use these questions to review the topics for Midterm 1. Work on separate paper, show your reasoning, and include units in applications. Give exact answers unless a decimal approximation is requested. This sheet is a collection of practice problems, not a timed practice exam.

- Quadratics and constraints.** Let $f(x) = x^2 - 6x + 5$.
 - Find the vertex and all x - and y -intercepts. Sketch the graph.
 - Find the maximum and minimum values of f on $[0, 5]$, and every x where each occurs.
 - Repeat part (b) on $[4, 7]$. Explain how the constraint changes the answer.
- Reading a graph.** The entire graph of g is shown below. Each piece is a line segment. Open circles are excluded and filled circles are included.



- Find $g(-4)$, $g(-1)$, $g(0)$, $g(2)$, $g(4)$, and $g(6)$, if defined.
- Give the domain and range in interval notation.
- Find the absolute maximum and minimum values, and where they occur.
- Give the open intervals on which g is increasing or decreasing.
- Find $\lim_{x \rightarrow -1} g(x)$, $\lim_{x \rightarrow 2^-} g(x)$, $\lim_{x \rightarrow 2^+} g(x)$, $\lim_{x \rightarrow 2} g(x)$, and $\lim_{x \rightarrow 6^-} g(x)$. (Write DNE if a limit does not exist.)
- Explain the difference between $g(2)$ and the one-sided limits at 2.
- Find $\lim_{x \rightarrow 4^-} g(x)$, $\lim_{x \rightarrow 4^+} g(x)$, and $\lim_{x \rightarrow 4} g(x)$. Compare these limits with $g(4)$.

- Domains.** Find the real domain of each function. Write your answers in interval notation.

$$(a) \frac{x+1}{x^2-9} \quad (b) \ln(7-2x) \quad (c) \sqrt{3x+6} \quad (d) \frac{1}{\sqrt{x-4}}.$$

- Cost, revenue, profit, and whole-number quantities.** A shop sells gift baskets. The price per basket is $p(q) = 52 - 3q$ dollars when q baskets are sold. The fixed cost is \$40 and each basket costs \$7 to make. All baskets produced are sold, and capacity is 12 baskets.
 - Write $C(q)$, $R(q)$, and $P(q)$. First use the real domain $[0, 12]$.
 - Find the vertex and all intercepts of the profit function. Sketch it on the stated domain.

- (c) Find the maximum and minimum profit on $[0, 12]$ and where each occurs.
- (d) Solve $P(q) = 0$ and interpret the solutions. For which real quantities is profit positive?
- (e) Now require $q \in \{0, 1, \dots, 12\}$. Find every quantity maximizing profit and the resulting profit. Which whole-number quantities earn a profit?
5. **Supply and demand.** At a price of \$10, customers demand 120 units per week; at a price of \$16, they demand 60 units per week. Assume a linear demand relationship. Suppliers offer q units per week at a price $p_S(q) = 4 + 0.05q$ dollars.
- (a) Write the demand price $p_D(q)$ as a function of quantity.
- (b) Find a reasonable real domain for p_D if both price and quantity must be nonnegative. Interpret its slope and vertical intercept.
- (c) Find the equilibrium quantity and price by solving $p_D(q) = p_S(q)$.
- (d) Write the revenue function using the demand price and find the quantity and price that maximize revenue on its real domain.
6. **Building and maximizing a piecewise model.** A business plans an advertising campaign. With no advertising, campaign profit is $-\$200$. For the first \$400 spent, each additional dollar of advertising increases profit by \$3. Beyond \$400, each additional dollar decreases profit by \$1. The largest permitted budget is \$1,000. Let x be the advertising budget in dollars and $P(x)$ the profit in dollars.
- (a) Write P as a piecewise function. What is the domain?
- (b) Graph P , labeling endpoints, intercepts, and the point where the formula changes.
- (c) Find the maximum profit and the budget that achieves it.
- (d) Find the average rate of change on $[200, 800]$ and interpret it.
- (e) Find the equation of the secant line through $(200, P(200))$ and $(800, P(800))$.
- (f) Evaluate the left-hand, right-hand, and two-sided limits at $x = 400$.
7. **Average rates of change and secant lines.** Let $f(x) = x^2 - 3x + 2$.
- (a) Find the average value of f on $[1, 4]$.
- (b) Find the average rate of change of f on $[1, 4]$. Explain the difference between your answer in this part and the previous part.
- (c) Find the equation of the secant line through $(1, f(1))$ and $(4, f(4))$.
- (d) Find and simplify $\frac{f(x+h) - f(x)}{h}$ for $h \neq 0$.
8. **Marginal cost, revenue, and profit.** A company sells a product and models cost and revenue, in dollars, by $C(q) = 100 + 6q + q^2$ and $R(q) = 30q$ for $0 \leq q \leq 12$, where q is the number of units sold.
- (a) Write the profit function $P(q)$. Evaluate $P(5)$ and $P(6)$, including units. Interpret the meaning of these numbers.
- (b) Let $M_C(h) = \frac{C(5+h) - C(5)}{h}$. Evaluate $M_C(5), M_C(3), M_C(1)$, interpret the meaning of each of these numbers. Are there reasonable units for $M_C(h)$?
- (c) Find the marginal cost, revenue and profit at $q = 5$. Include units and interpret the meaning of each of these numbers.

9. Exponential models and thresholds.

- (a) A machine initially worth \$12,000 loses 15% of its value each year. Write an exponential model $V(t)$ for real $t \geq 0$. Find and interpret $V(2)$.
- (b) Find the average rate of change of V on $[0, 2]$ and interpret it.
- (c) Solve $V(t) = 6000$. For which real times is $V(t) < 6000$? Give an exact threshold and an approximation to two decimal places.
- (d) A subscriber base starts at 800 and grows by 12% each year. Write a model $N(t)$. Find all real times when it exceeds 1,500 and the first whole-year anniversary at which it exceeds 1,500.

10. Exponential and logarithmic equations. Solve each equation over the real numbers. Check all logarithm arguments and give exact answers.

$$\begin{array}{ll} \text{(a)} \quad 3^{2x-1} = 20 & \text{(b)} \quad 500e^{-0.2t} = 125 \\ \text{(c)} \quad \ln(x-1) + \ln(x+1) = \ln 8 & \text{(d)} \quad \log_2(x-3) = 4. \end{array}$$

11. Basic limits and limit laws. Suppose $\lim_{x \rightarrow a} f(x) = 4$ and $\lim_{x \rightarrow a} g(x) = -2$. Consider the following limits.

- (a) Evaluate each of the limits (write DNE if it does not exist)

$$\lim_{x \rightarrow a} f(x) + g(x), \quad \lim_{x \rightarrow a} 3f(x), \quad \lim_{x \rightarrow a} f(x)g(x), \quad \lim_{x \rightarrow a} f(x)/g(x), \quad \lim_{x \rightarrow a} f(x)^2, \quad \sqrt[3]{g}.$$

- (b) Determine if each of the limits are finite, infinite, do not exist, or if there is not enough information to determine.

$$\lim_{x \rightarrow a} \frac{g(x) - 2}{4 - g(x)^2}, \quad \lim_{x \rightarrow a} \frac{f(x)^2 - 4}{g(x)}, \quad \frac{g(x)^2 + f(x)}{f(x)}, \quad \frac{g(x)^2}{f(x) - 4}.$$

12. Evaluate the following limits:

$$\begin{array}{llll} \text{(a)} \quad \lim_{x \rightarrow 2} (3x^2 - x + 4) & \text{(b)} \quad \lim_{x \rightarrow 5} \sqrt{x+4} & \text{(c)} \quad \lim_{x \rightarrow 0} 2^x & \text{(d)} \quad \lim_{x \rightarrow e} \ln x. \end{array}$$

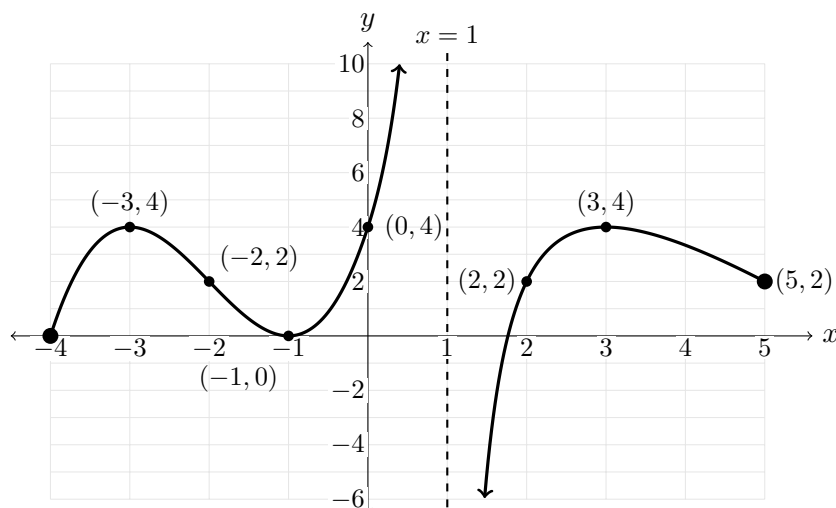
13. Rational-function limits. Evaluate each limit. If a two-sided limit does not exist, explain using one-sided behavior. Distinguish a function value from a limit.

$$\begin{array}{llll} \text{(a)} \quad \lim_{x \rightarrow 2} \frac{x^2 + 3}{x + 1}, & \text{(b)} \quad \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}, & \text{(c)} \quad \lim_{x \rightarrow -2} \frac{x^2 + x - 2}{x + 2}, & \text{(d)} \quad \lim_{x \rightarrow 1} \frac{1}{x - 1}. \end{array}$$

$$\text{14. Piecewise functions and limits. Let } F(x) = \begin{cases} x + 2, & x < 1, \\ 5, & x = 1, \\ 4 - x, & 1 < x \leq 3, \\ 2x - 7, & x > 3. \end{cases}$$

- (a) Graph F , using open and filled circles correctly. Find $F(1)$ and $F(3)$.
- (b) Find both one-sided limits and the two-sided limit at $x = 1$ and at $x = 3$.
- (c) If only $F(1)$ is changed, what value makes it equal to $\lim_{x \rightarrow 1} F(x)$? Can changing only $F(3)$ make the two-sided limit at 3 exist? Explain.

15. **Reading a more complicated graph.** The graph of H has domain $[-4, 1) \cup (1, 5]$. The dashed line is a vertical asymptote at $x = 1$. Arrows show that the curves continue without bound as they approach the asymptote. The filled endpoints are included. Use the graph to answer the questions; no formula is needed.



- (a) Give the open intervals on which H is increasing and those on which it is decreasing.
- (b) Give the open intervals on which H is concave up and those on which it is concave down. Identify each inflection point by its coordinates.
- (c) On each interval below, find the absolute maximum and absolute minimum values of H and *every* x where each occurs. All these intervals lie entirely in the domain.

$$[-4, -1], \quad [-2, 0], \quad [-1, 0], \quad [2, 5], \quad [2, 3].$$

- (d) Can we maximize/minimize H on $[-1, 2]$?
- (e) Evaluate the following limits. Use $+\infty$ or $-\infty$ for unbounded one-sided behavior, and explain if a two-sided limit does not exist.

$$\lim_{x \rightarrow 1^-} H(x), \quad \lim_{x \rightarrow 1^+} H(x), \quad \lim_{x \rightarrow 1} H(x), \quad \lim_{x \rightarrow -2} H(x), \quad \lim_{x \rightarrow 3} H(x).$$